

TANVIR HOSSAIN

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RESEARCH INTERESTS

My research develops attacks and defenses for trustworthy microelectronics across the hardware and software stack. I work on power and electromagnetic side-channel analysis, fault injection, hardware Trojan detection and mitigation, and intellectual-property protection, combining digital VLSI and ASIC design flows (RTL design, verification, and synthesis) with machine learning to build golden-chip-free, pre-deployment security evaluation and assurance techniques for integrated circuits.

EDUCATION

Doctor of Philosophy (Ph.D.) in Electrical Engineering, The University of Kansas Expected 2026
Advisor: Dr. Tamzidul Hoque, Assistant Professor, EECS, KU.

Master of Science (M.S.) in Electrical Engineering, The University of Kansas May 2025
Advisor: Dr. Tamzidul Hoque, Assistant Professor, EECS, KU.

Bachelor of Science in Electrical and Electronic Engineering
Ahsanullah University of Science and Technology, Bangladesh, CGPA: 3.69 2016 - 2021

PUBLICATIONS

25 papers total: 18 peer-reviewed publications (8 journal articles, 10 conference papers) and 7 manuscripts under review. Name in **bold**; joint first authorship noted where applicable.

Journal Articles:

1. **Tanvir Hossain**, Andrea Ramirez-Salgado, Ertewaa Alsahayan, Swarup Bhunia, Pavlo Antonenko, and Tamzidul Hoque, “**Game On: Gamifying Computer Hardware Fundamentals Learning Using a Configurable Platform**,” *IEEE Transactions on Education*, 2026. DOI: [10.1109/TE.2026.3728763](https://doi.org/10.1109/TE.2026.3728763). (Accepted for publication, forthcoming online)
2. **Tanvir Hossain**, S. M. Mojahidul Ahsan, Prabuddha Chakraborty, Swarup Bhunia, and Tamzidul Hoque, “**Attacking the Protections of Hardware using Trojan-Assisted Meta Attacks**,” *ACM Transactions on Design Automation of Electronic Systems*, 2026. DOI: [10.1145/3804452](https://doi.org/10.1145/3804452).
3. **Tanvir Hossain**, Matthew Showers, Mahmudul Hasan, and Tamzidul Hoque, “**On the Security of Lightweight Homomorphic Obfuscation for Protecting Against Hardware Trojans**,” *IACR Transactions on Cryptographic Hardware and Embedded Systems*, vol. 2026, no. 2, pp. 736–763, 2026. DOI: [10.46586/tches.v2026.i2.736-763](https://doi.org/10.46586/tches.v2026.i2.736-763).
4. **Tanvir Hossain**, Ashutosh Ghimire (joint first author), Fathi Amsaad, Tamzidul Hoque, and Ahmed Sherif, “**AI-Assisted Side-Channel Analysis for Golden-Chip Free Detection and Localization of Hardware Trojans**,” *Journal of Hardware and Systems Security*, 2026. DOI: [10.1007/s41635-026-00182-4](https://doi.org/10.1007/s41635-026-00182-4). (Accepted for publication, forthcoming online)
5. Sriram Prabhakara Rao, Shree Ram Abayankar Balaji, Farishta Rahman, Tasneem Suha, Tanzim Mahfuz, **Tanvir Hossain**, Mahmudul Hasan, Md Sakib Hasan, Tamzidul Hoque, Prabuddha Chakraborty, and Prakash Ranganathan, “**Secure and Trustworthy Microelectronics: Vulnerabilities, Solutions, and Trends**,” *Journal of Hardware and Systems Security*, vol. 9, pp. 163–189, 2025. DOI: [10.1007/s41635-025-00165-x](https://doi.org/10.1007/s41635-025-00165-x).
6. Md Mashfiq Rizvee, Fairuz Shadmani Shishir, **Tanvir Hossain**, Tamzidul Hoque, Domenic Forte, and Sumaiya Shomaji, “**A Persistent Hierarchical Bloom Filter-based Framework for Scalable Authentication and Tracking of ICs**,” *ACM Journal on Emerging Technologies in Computing Systems*, vol. 21, no. 4, pp. 1–22, 2025. DOI: [10.1145/3748650](https://doi.org/10.1145/3748650).
7. **Tanvir Hossain** and Md Istiaque Rahaman, “**Band gap tuning and variable quantum barrier of armchair graphene nanoribbons by inducing antidote topologies**,” *Journal of Computational Electronics*, vol. 23, no. 1, pp. 32–41, 2024. DOI: [10.1007/s10825-023-02108-7](https://doi.org/10.1007/s10825-023-02108-7).
8. **Tanvir Hossain**, Md Istiaque Rahaman, and Mahbub Alam, “**Antidote-induced armchair graphene nanoribbon based resonant tunneling diodes**,” *Semiconductor Science and Technology*, vol. 36, no. 7, article 075023, 2021. DOI: [10.1088/1361-6641/ac01fc](https://doi.org/10.1088/1361-6641/ac01fc).

Conference Proceedings:

1. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “**SPHINX: A Framework for Security Primitive Hardware Identification and Extraction**,” *GLSVLSI '26: Great Lakes Symposium on VLSI 2026*, Canandaigua, NY, USA, June 2026. DOI: [10.1145/3787109.3815307](https://doi.org/10.1145/3787109.3815307). (Accepted for publication, forthcoming online)

2. Andrea Ramirez-Salgado, Swarup Bhunia, Pavlo Antonenko, Woorin Hwang, Christine Wusylko, Yessy Ambarwati, **Tanvir Hossain**, Tamzidul Hoque, and Rohan Kalavakonda, “**BOARD# 278: NSF IUOE: Empowering Future Engineers. An Inclusive Curriculum for AIoT and Intelligent Embedded Systems,**” *2025 ASEE Annual Conference & Exposition*, 2025. DOI: [10.18260/1-2--55641](https://doi.org/10.18260/1-2--55641).
3. S. M. Mojahidul Ahsan, Muhammad Sakib Shahriar, Mrityika Chowdhury, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “**Accurate, Yet Scalable: A SPICE-based Design and Optimization Framework for eNVM based Analog In-memory Computing,**” *Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, pp. 1–9, 2024. DOI: [10.1145/3676536.3676827](https://doi.org/10.1145/3676536.3676827).
4. Andrea Ramirez-Salgado, **Tanvir Hossain**, Swarup Bhunia, and Pavlo Antonenko, “**Board 393: Supporting Hardware Engineering Career Choice in First-Year Engineering Students,**” *2024 ASEE Annual Conference & Exposition*, 2024. DOI: [10.18260/1-2--46979](https://doi.org/10.18260/1-2--46979).
5. Ashutosh Ghimire, Fathi Amsaad, **Tanvir Hossain**, Tamzidul Hoque, and Ahmed Sherif, “**FPGA Hardware Trojan Detection: Golden-Free Machine Learning Approach,**” *NAECON 2023 - IEEE National Aerospace and Electronics Conference*, pp. 181–186, 2023. DOI: [10.1109/NAECON58068.2023.10365812](https://doi.org/10.1109/NAECON58068.2023.10365812).
6. Fairuz Shadmani Shishir, Md Mashfiq Rizvee, **Tanvir Hossain**, Tamzidul Hoque, Domenic Forte, and Sumaiya Shomaji, “**A Persistent Hierarchical Bloom Filter-based Framework for Authentication and Tracking of ICs,**” *3rd IEEE International Workshop on Silicon Lifecycle Management*, 2024. DOI: [10.48550/arXiv.2408.16950](https://doi.org/10.48550/arXiv.2408.16950).
7. Andrea Ramirez-Salgado, **Tanvir Hossain**, Tamzidul Hoque, Swarup Bhunia, Mary Jo Koroly, Bradford Davey, and Pavlo Antonenko, “**Board 265: Engaging Students in Exploring Computer Hardware Fundamentals Using FPGA Board Games,**” *2023 ASEE Annual Conference & Exposition*, 2023. DOI: [10.18260/1-2--42715](https://doi.org/10.18260/1-2--42715).
8. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “**Potential and Pitfalls of Multi-valued Logic Circuits for Hardware Security,**” *2023 IEEE 16th Dallas Circuits and Systems Conference (DCAS)*, pp. 1–6, 2023. DOI: [10.1109/DCAS57389.2023.10130261](https://doi.org/10.1109/DCAS57389.2023.10130261).
9. S. M. Mojahidul Ahsan, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “**Resistive RAM-based PUF: Challenges and Opportunities,**” *2023 IEEE 16th Dallas Circuits and Systems Conference (DCAS)*, pp. 1–6, 2023. DOI: [10.1109/DCAS57389.2023.10130179](https://doi.org/10.1109/DCAS57389.2023.10130179).
10. **Tanvir Hossain**, Md Shyeem Rahman, Md Mushfiqur Rahman, and Ahmed Rabbi Arini Dibbo, “**Performance Analysis of Graphene Nanoribbon Field Effect Transistor (GNRFET) based 6T and 7T SRAMs,**” *2020 23rd International Conference on Computer and Information Technology (ICCIT)*, pp. 1–5, 2020. DOI: [10.1109/ICCIT51783.2020.9392713](https://doi.org/10.1109/ICCIT51783.2020.9392713). (**Best AUST paper award**)

Under Review:

1. **Tanvir Hossain**, Abhinav Mahadevan, Jasper Van Woudenberg, Rajesh Velegalati, and Arindam Bhattacharyya, “**Glitch-Lab: A Hardware-in-the-Loop Optimizer for Physical Fault Injection,**” *arXiv preprint*, arXiv:2609.00502 [cs.CR], 2026. DOI: [10.48550/arXiv.2609.00502](https://doi.org/10.48550/arXiv.2609.00502). (Under review)
2. **Tanvir Hossain**, Sudipta Paria, Swarup Bhunia, and Tamzidul Hoque, “**REFORGE: LLM-Assisted RTL Bug Repair with Structured Context and Verification Feedback,**” *Design, Automation and Test in Europe (DATE) Conference 2027*, 2027. (Under review)
3. Ertewaa Alsahayan, Alex Manley, **Tanvir Hossain**, Mohammad Alian, and Tamzidul Hoque, “**Toward Reliable LLM-Assisted Design Space Exploration Under Multi-Performance and Multi-Cost Constraints,**” *Microprocessors and Microsystems: Embedded Hardware Design (MICPRO)*, 2026. (Major revision)
4. Rajesh Velegalati, Jasper Van Woudenberg, **Tanvir Hossain**, Abhinav Mahadevan, and Arindam Bhattacharyya, “**From Structured Search to LLM Operators: Automating Physical Fault-Injection Campaigns,**” *GOMACTech (Government Microcircuit Applications & Critical Technology) Conference*, 2027. (Under review)
5. Ertewaa Saud Alsahayan, **Tanvir Hossain**, and Tamzidul Hoque, “**REMODEL-V: Reliable LLM-Guided Design Space Exploration for Soft-Error-Aware Microarchitectures,**” *GOMACTech (Government Microcircuit Applications & Critical Technology) Conference*, 2027. (Under review)
6. S. M. Mojahidul Ahsan, Mohammad Sakib Shahriar, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “**SPICECiM-X: An Accurate Circuit-Level Design and Optimization Framework for Analog In-Memory Transformer Inference,**” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2026. (Under review)
7. **Tanvir Hossain**, Mahmudul Hasan, Jackie Ho, Garrett Rose, Tamzidul Hoque, and Md Sakib Hasan, “**PURL: Physically Unclonable and Reconfigurable Logic Locking,**” *Journal of Hardware and Systems Security*, 2026. (Under review)

RESEARCH EXPERIENCE

The University of Kansas

Graduate Research Assistant, Hardware Security (Advisor: Dr. Tamzidul Hoque)

Jan 2022 – Present
Lawrence, KS

- Conduct research on hardware security spanning side-channel analysis, fault injection, and hardware Trojan detection and mitigation, with results published in IACR TCHES, ACM TODAES, and the Journal of Hardware and Systems Security.
- Develop golden-chip-free and machine-learning-assisted security evaluation methods for integrated circuits, collaborating with researchers across multiple universities.

Keysight Technologies

Research Intern, Machine Learning Security (Applied ML & Fault Injection)

May 2026 – Dec 2026
Santa Rosa, CA

- Developing model-guided optimization and adaptive-learning workflows for fault-injection parameter tuning, including experiment design, telemetry instrumentation, scalable hyperparameter search, and reliability metrics.
- Delivering a prototype demonstration targeting improved vulnerability-detection coverage over baseline flows.

NSF Research Experience for Undergraduates (REU)

Graduate Research Mentor

Summer 2025

- Mentored four undergraduate researchers in a collaborative NSF REU program between the University of Kansas and Wright State University.

SELECTED RESEARCH PROJECTS

- Trojan-Assisted Meta-Attacks Against Design-for-Security Primitives:** Developed a unified framework to evaluate how hardware Trojans can systematically identify and neutralize Design-for-Security (DfS) primitives in zero-trust microelectronics supply chains. Demonstrated attacks on Ring Oscillator Physically Unclonable Functions (RO-PUFs) and Dynamically Obfuscated Scan Chains (DOSC), enabling authentication bypass, scan-protection removal, and motivating Trojan-aware countermeasures.
- Runtime Protection of Secrets in Untrusted COTS Processors:** Developed HOACS, a software-based framework that uses Residue Number Coding (RNC) to keep sensitive data encoded during execution on commercial off-the-shelf (COTS) processors. Implemented both C-library and LLVM-based transformations to protect cryptographic and arithmetic workloads from hardware-Trojan leakage and side-channel attacks without requiring hardware modification.
- Side-Channel-Driven Security Evaluation of Opaque COTS Hardware:** Proposed a gray-box evaluation framework for assessing unknown COTS devices before deployment when internal design information is unavailable. The approach leverages power and electromagnetic (EM) side-channel measurements to identify anomalous behaviors, guide vulnerability discovery, and support pre-deployment assurance under zero-trust hardware assumptions.
- Gamified Learning Framework for Computer Hardware Fundamentals:** Developed hardware-based games to teach fundamental concepts of computer hardware using Verilog, FPGA, and ESP32 development boards. *Funded by NSF, 2022 to present.*

GRANTS & FUNDING

Contributed as a graduate research assistant to the following federally funded projects.

MARCH 2022 - PRESENT	Collaborative Research: Gamified Learning of Computing Hardware Fundamentals Funding Agency: National Science Foundation, Award Number: 2142696
OCTOBER 2024 - PRESENT	Collaborative Research: Engaging High Schoolers in Integrated Computer Science and Engineering Through Hands-On Experiences with Microelectronics and Artificial Intelligence Funding Agency: National Science Foundation, Award Number: 2405375
2024 - PRESENT	Collaborative Research: SaTC: CORE: Medium: KIPPER: A Scalable Learning-Guided Hardware IP Protection Platform Funding Agency: National Science Foundation, Award Number: 2350364
MARCH 2026 - PRESENT	Collaborative Research: REU Site: TRUST: Training Research for Undergraduate Students in Secure and Trusted Systems Funding Agency: National Science Foundation, Award Number: 2349629

HONORS & AWARDS

MAY 2026	National Science Foundation (NSF) Travel Grant IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2026 Symposium)
JANUARY 2024	First Place (Poster Presentation) In the I2S Student Research Symposium
MAY 2023	Champion: Microelectronics Security Challenge (IP security) IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2023 Symposium)
MAY 2023	National Science Foundation (NSF) Travel Grant IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2023 Symposium)
APRIL 2023	David D. and Mildred H. ROBB Award Awarded by the EECS department of the University of Kansas
APRIL 2023	People's Choice Award (Poster Presentation) Among 16 participants in the 2023 Research Showcase by the Graduate Engineering Association of KU
APRIL 2023	Third Place (Poster Presentation) Among 16 participants in the 2023 Research Showcase by the Graduate Engineering Association of KU
MARCH 2023	Second Place (Research Showcasing) Among 20 participants in the I2S Student Research Symposium
DECEMBER 2022	Bangladesh Sweden Trust Fund (A travel grant for Graduate study in USA)
JUNE 2022	National Science Foundation (NSF) Travel Grant IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2022 Symposium)
MARCH 2021	Best AUST Paper Award The 23rd International Conference on Computer and Information Technology (ICCIT 2020)
FALL 2019	Tuition Waiver Award (Based on the result of the top 8% students)
AUGUST 2018	Runner-up Hardware Exhibition, Inter University Software and Hardware Carnival-2018, University of Asia Pacific
APRIL 2018	Second Runner-up Project Showcasing of Technival '18, Khulna University of Engineering and Technology (KUET)
FEBRUARY 2018	Second Runner-up Hardware of Engenius '18, Organized by AUST EEE Society
FEBRUARY 2018	Champion Hardware Showcasing of Technomania '18, AUST CSE Society

TEACHING & OUTREACH

The University of Kansas

Aug 2024 – Dec 2024

Graduate Teaching Assistant, EECS

Lawrence, KS

- Led lab sessions and graded for undergraduate digital logic and computer architecture, contributing to course materials and exams.
- Guided project-based learning using the MARS (MIPS) and gem5 (x86) simulators.

The University of Kansas

Fall 2023

Co-Instructor

Lawrence, KS

- Co-taught “Hands-On Introduction to Computer Hardware using FPGA and Microcontroller,” developing modules with FPGAs, simulations, and the ESP32 Aha development board.

The University of Kansas

Summer 2023, 2024

Summer Camp Instructor, EECS

Lawrence, KS

- Designed and ran gamified, hands-on summer camps that teach high-school students computer hardware fundamentals using FPGAs and basic electronics.

INVITED TALKS & PRESENTATIONS

JANUARY 2025	Leakage Unplugged: A Hands-On Tutorial on Power Side-Channel Attacks Research Workshop in the I2S Student Research Symposium 2025
FEBRUARY 2024	Power Side-Channel and Fault Attacks on ASCON and their Countermeasures Information Advantage Scholars Visit at I2S, KU

PROFESSIONAL SERVICE

JOURNAL REVIEWER	SN Computer Science Physica Scripta Journal of Physics: Condensed Matter IEEE Embedded Systems Letters Journal of Hardware and Systems Security
CONFERENCE REVIEWER	Technical Program Committee (TPC): IEEE Computer Society Annual Symposium on VLSI (ISVLSI 2026) 2026 ASEE Annual Conference & Exposition 2024 ASEE Annual Conference & Exposition
OTHER RESPONSIBILITIES	Judge: I2S Undergraduate Student Research Showcase

TECHNICAL SKILLS

Hardware Description Language:	Verilog, VHDL.
FPGA Design Tools:	Xilinx Vivado, ISE, SDK; Quartus.
Programming Languages:	Python, C, MATLAB, Assembly (MIPS).
Design & Verification Tools:	Cadence Virtuoso, Genus, Encounter; ModelSim; PSpice, HSPICE.
Simulation and Analysis Tools:	gem5, Intel VTune, McPAT.
Hardware Platforms:	FPGA Boards: HaHa Board (Intel MAX10), Altera DE2-115 FPGA, Basys 3 Artix-7, Nexys A7-100T. Embedded Systems: Raspberry Pi, Arduino, STM32.
Compilers and Toolchains:	LLVM.
Side-Channel Analysis:	Keysight Spyder, Inspector, Pinata, high-precision EM probe; ChipWhisperer Husky; PicoScope; Tektronix DPO70404C Oscilloscope; Analog Discovery 2.
Scripting:	Tcl (Tickle), Bash, Git.

SIGNIFICANT COURSES

MAJOR COURSES IN B.SC.:	VLSI - I, VLSI - II, Analog Integrated Circuit, Processing and Fabrication Technology, Computer Architecture, Compound Semiconductor and Hetero-Junction Devices.
MAJOR COURSES IN PH.D. :	Intro. Hardware Security, Advance Software Security, Modern Computer Architecture, Machine Learning, Embedded Machine Learning, Advance Data Science, Computer Vision.